

Math 220: Mathematical Reasoning and Proof
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All Solutions

Last updated: August 28, 2026

1	(September 4): Introduction to course. First proof.	2
2	(September 11): Logic. “Direct” proofs	3
3	(September 18): More “Direct” proofs; proof by cases	7
4	(September 25): Proof by contradiction	11
5	(October 2): Induction	15
6	(October 16): Set theory. Basic operations. Proofs with sets	20
7	(October 23): More sets. DeMorgans laws. Cartesian Products. Power sets	25
8	(October 30): Introduction to functions; images and surjectivity	36
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10	(November 20): More injectivity; composition of functions	59
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12	(December 9): Relations; (Un)countability	77